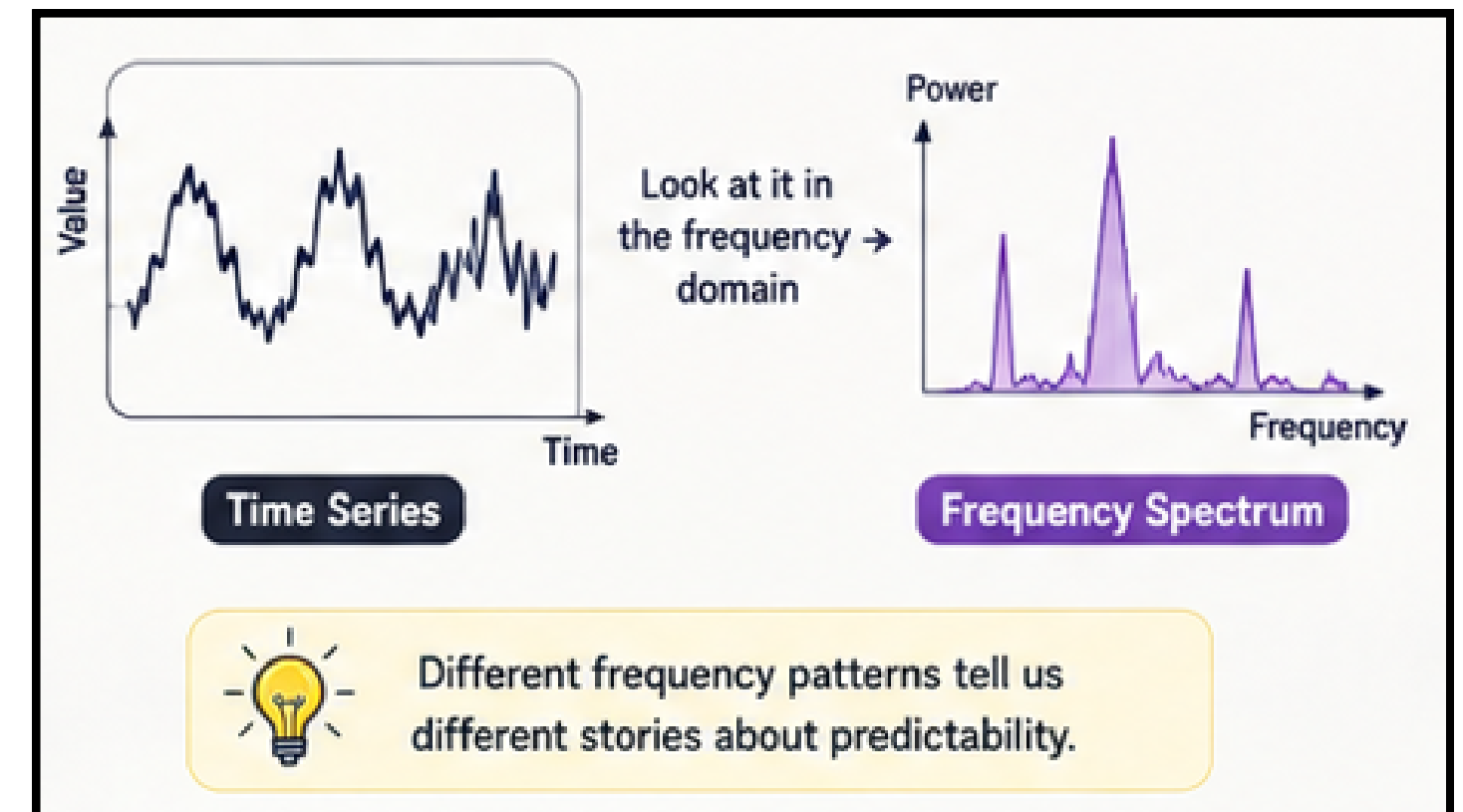


Is your data hiding a pattern?

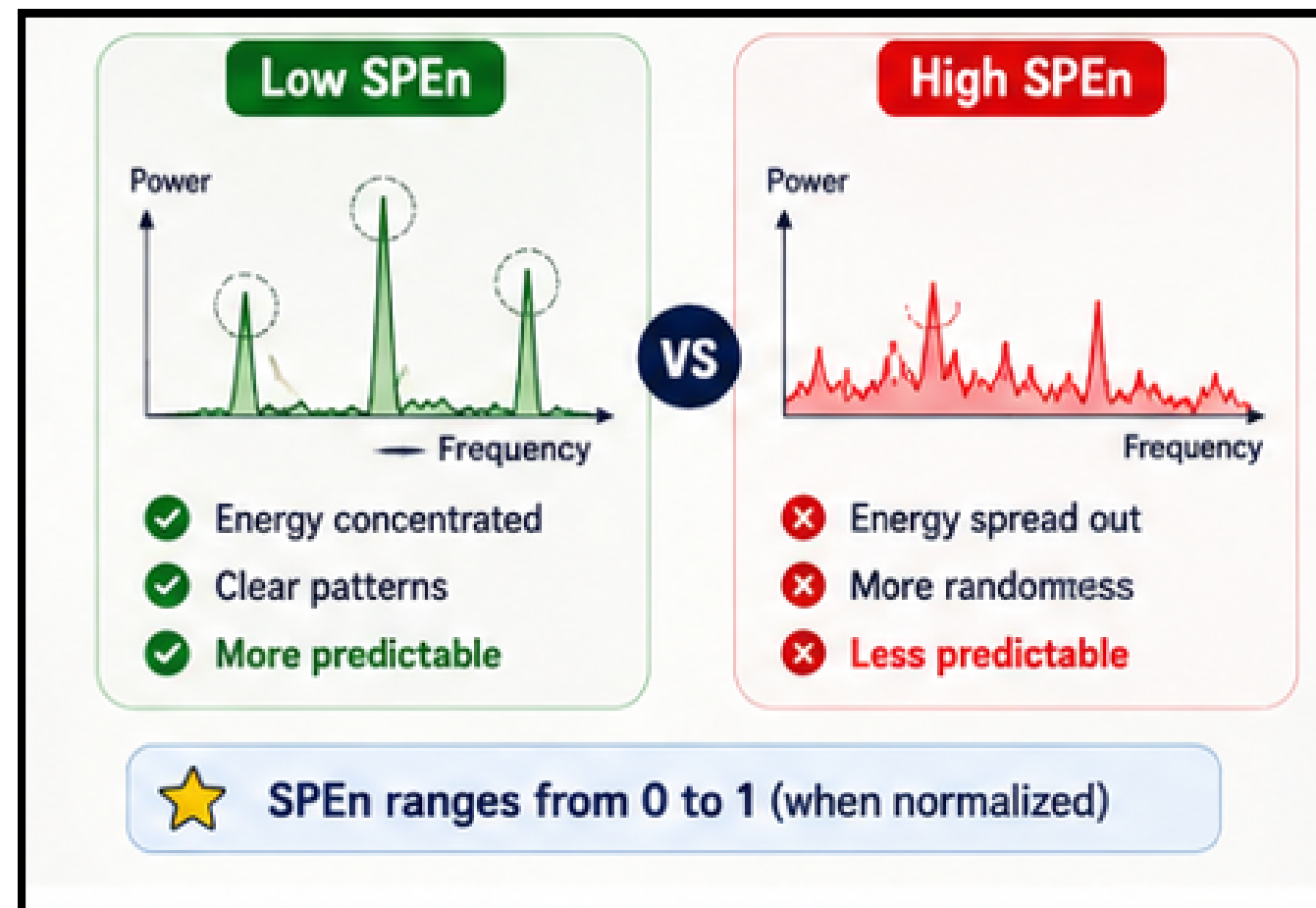
- Not every time series is equally predictable.
- Some have clear, concentrated patterns.
- Others have energy scattered across many frequencies.
- **Spectral Entropy (SpEn)** helps us understand how **predictable** a time series is by looking at its **frequency structure**.



What is SPEn?

SPEn = Spectral Entropy

It measures how spread out the signal's energy is across frequencies.



How SpEn is calculated?

1. Convert the series to frequency using FFT
2. Compute the power spectrum(energy at each frequency)
3. Normalize the power to get probabilities:

$$p_i = \frac{P_i}{\sum P_i}$$

4. Calculate Spectral Entropy

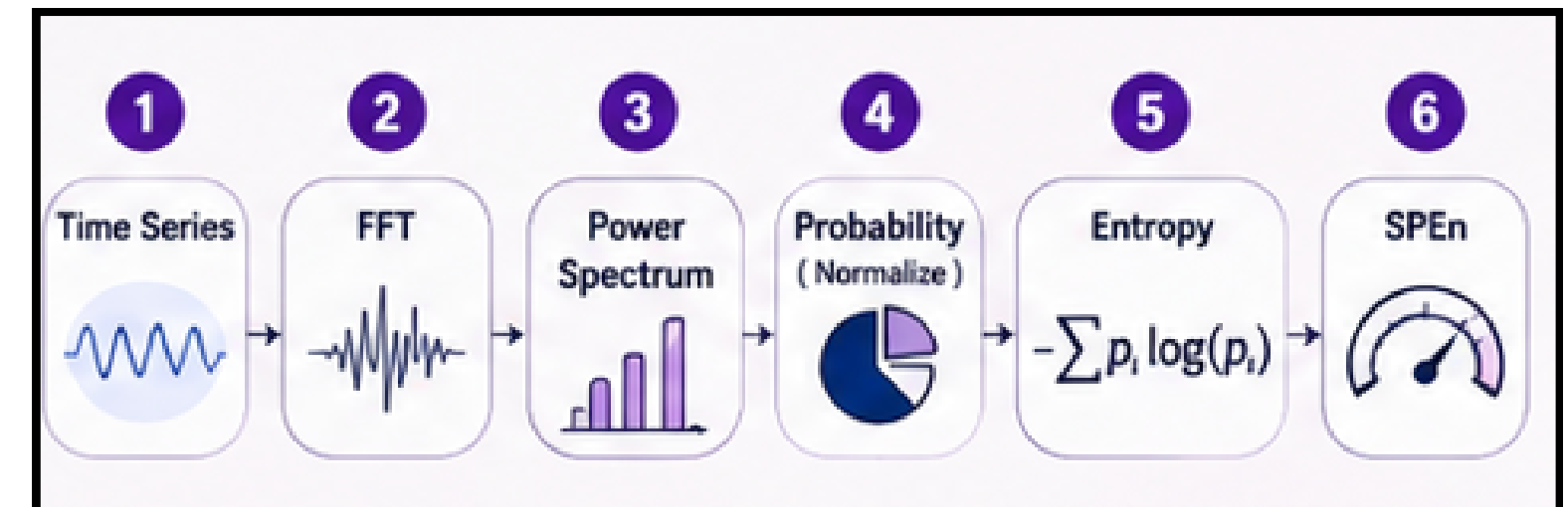
$$H = - \sum p_i \log(p_i)$$

5. Normalize the Entropy

$$SPEn = \frac{H}{\log(N)}$$

N = number of frequency bins

6. SpEn → 0 : Energy Concentrated
SpEn → 1 : Energy Widely distributed



REAL-WORLD EXAMPLE

Imagine we are taking two datasets here:

Electricity demand

Clear daily/weekly patterns

→ Energy concentrated around certain frequencies

→ Lower SPEn

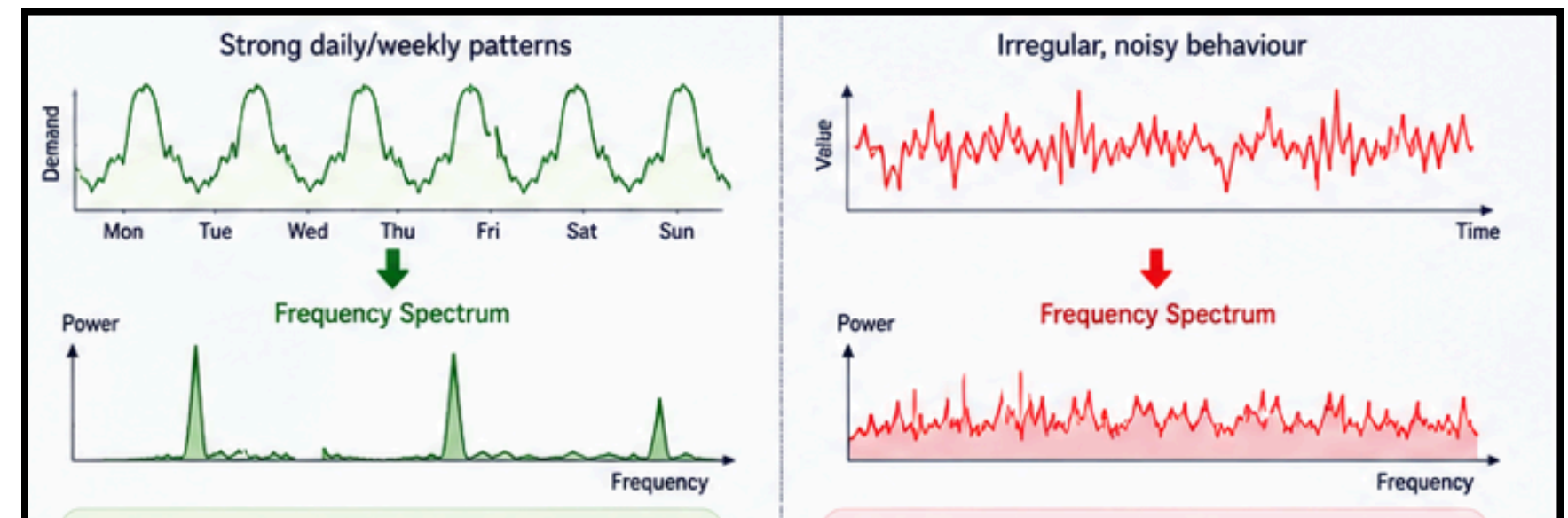
→ More forecastable

Noisy signal

Energy spread across many frequencies

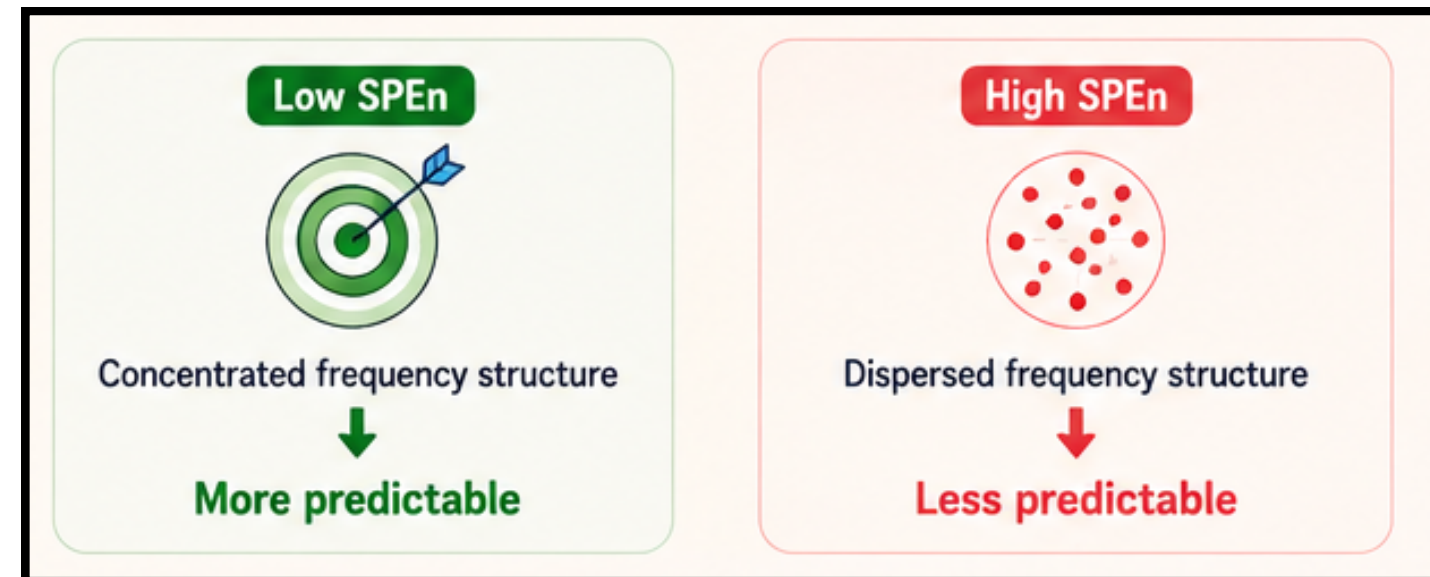
→ Higher SPEn

→ Less forecastable



TAKEAWAY

SpEn gives us a clue about Forecastability



Before Choosing a **forecasting** model, understand your **data** first